



Lab: Basic Knowledge of Equipment

1.1 Digital Multimeter (DMM):

The digital multimeter (DMM) is one of the most useful devices to measure voltage, current and resistance. Most DMMs have three terminals and two probes.

- (i) One black terminal - zero potential/ Ground
- (ii) One red terminal - for measuring voltage
- (iii) Two red terminals - for measuring current

One probe is continuously connected to the black terminal and another probe connects to one of the two red terminals depending on the measurement mode. Some advanced DMMs can also measure capacitance, inductance, detect terminals of transistors, diodes, etc.

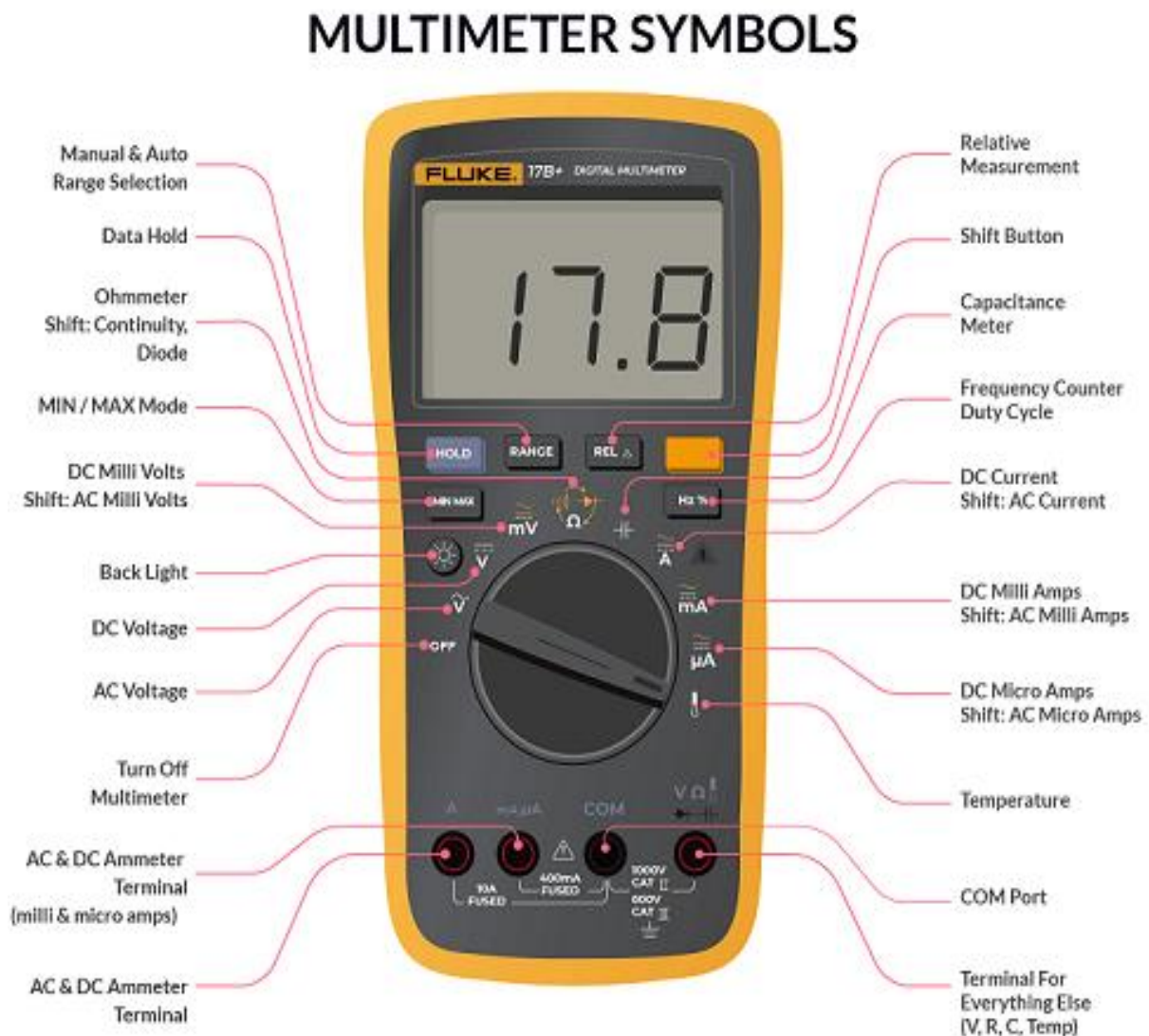


Fig. 1: Digital Multimeter

PRECAUTION

To avoid **damage of the DMM**:

- Keep it switched off while not in use.
- Before connecting the DMM, the measurement mode must be selected and its meter range should be placed to its highest value.
- The red probe must be connected to the correct terminal.

1.2 Voltage Measurement

Voltage is measured across the circuit elements / components. That is - a parallel connection is made with DMM and the desired element. Voltage measurement requires negative and positive polarity consideration. If the reading gives a positive value the polarity consideration is correct.

1.3 Current Measurement

Current is measured through the circuit components. So, current measurement requires series connection with the DMM. Current measurement also requires polarity consideration. Similar to voltage measurement a positive reading will indicate right current flow consideration.

1.4 Resistance Measurement

Resistances are the simplest form of circuit components. Commercially resistors come in many shapes, sizes. Most common types of resistors are color-coded carbon composition or carbon film resistors. Color codes are multi-colored bands that determine the resistor's value and tolerance. To measure the resistance two probes of DMM are connected to the two ends of the resistor. Again, resistance mode (Ohmmeter) must be selected before starting measurement.

PRECAUTION

Do **not** connect an **Ohmmeter to a live circuit**. Only connect the component of which the resistance is to be measured.

Another way of measuring resistance is reading color codes (printed colored rings) on the resistors.

Resistors Color Code:

Resistors use colored painted bands to indicate both their resistive value and their tolerance with the physical size of the resistor indicating its wattage rating. These colored painted bands produce a system of identification generally known as a **Resistors Color Code**.

An international and universally accepted resistor color code scheme was developed many years ago as a simple and quick way of identifying a resistor's ohmic value no matter what its size or condition. It consists of a set of individual-colored rings or bands in spectral order representing each digit of the resistor's value. The resistor color code markings are always read one band at a time starting from the left to the right, with the larger width tolerance band oriented to the right side indicating its tolerance. By matching the color of the first band with its associated number in the digit column of the color chart below the first digit is identified and this represents the first digit of the resistance value.



Fig. 2: 4 Bands Resistor

Again, by matching the color of the second band with its associated number in the digit column of the color chart we get the second digit of the resistance value and so on. The fourth and fifth bands are used to determine the percentage tolerance of the resistor. Resistor tolerance is a measure of the resistors variation from the specified resistive value and is a consequence of the manufacturing process and is expressed as a percentage of its “nominal” or preferred value. The resistor color code is read from left to right as illustrated below:

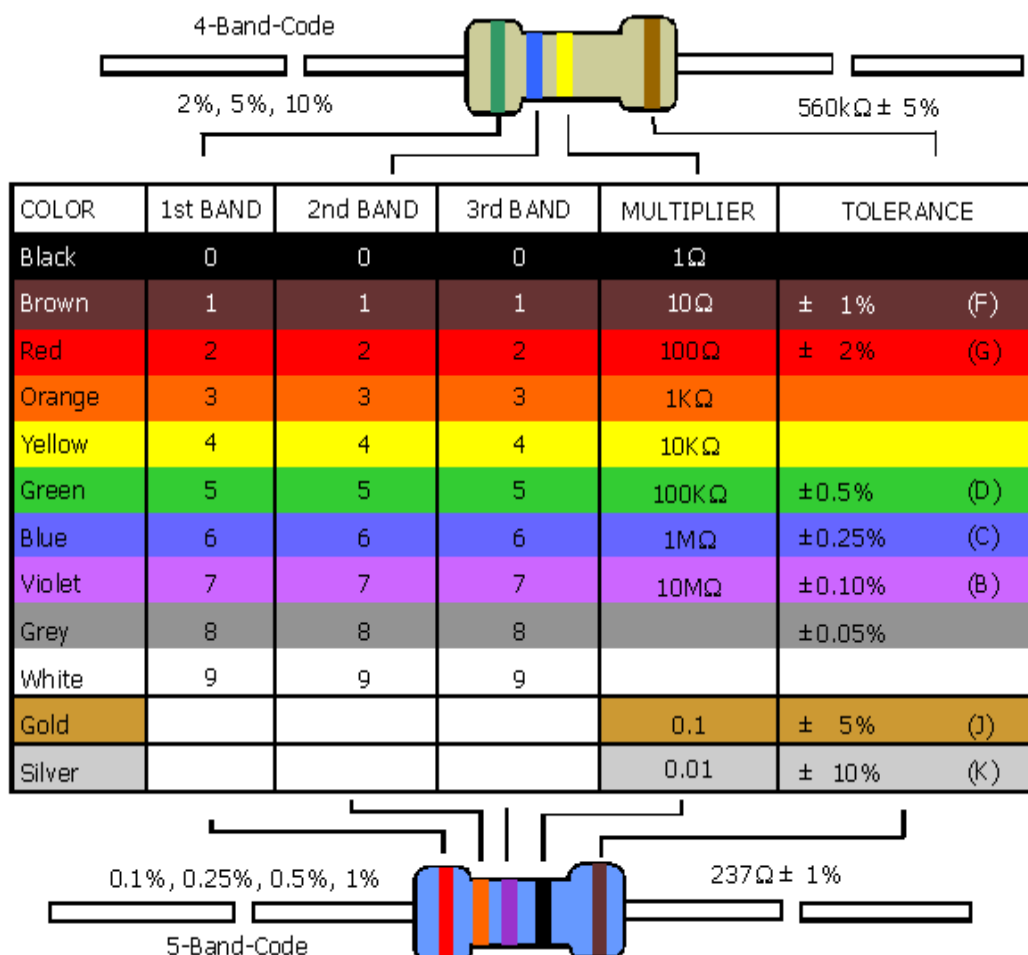


Fig. 3: The Standard Resistor Color Code Chart

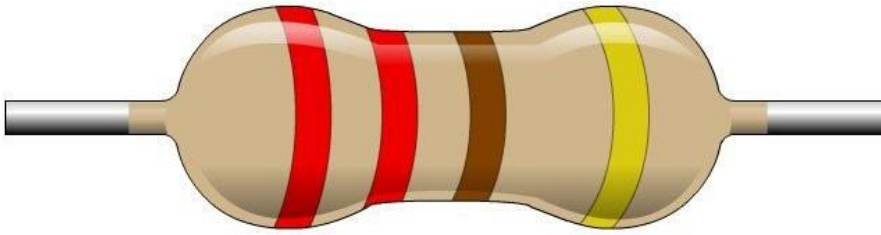
The **Resistor Color Code** system is all well and good, but we need to understand how to apply it in order to get the correct value of the resistor. The “left-hand” or the most significant colored band is the band which is nearest to a connecting lead with the color-coded bands being read from left-to-right as follows:

Digit, Digit, Multiplier $\pm$ tolerance = Color, Color, Multiplier in Ohm’s (Ω) $\pm$ tolerance

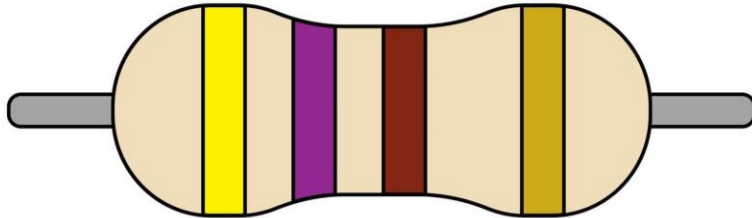
For example, a resistor has the following-colored markings:

Yellow, Violet, Red, Golden = $(4\ 7 \times 100\Omega \pm 5\%) = (4700\Omega \pm 5\%)$ or $(4.7\ \text{k}\Omega \pm 5\%)$

Test 1:



Test 2:



Test 3:



Test 4:



1.5 Breadboard (<http://wiring.org.co/learning/tutorials/breadboard/>)

A breadboard is a solder less device for temporary prototype with electronics and test circuit designs. Most electronic components in electronic circuits can be interconnected by inserting their leads or terminals into the holes and then making connections through wires where appropriate. The breadboard has strips of metal underneath the board and connects the holes on the top of the board. The metal strips are laid out as shown below. Note that the top and bottom rows of holes are connected horizontally and split in the middle while the remaining holes are connected vertically.

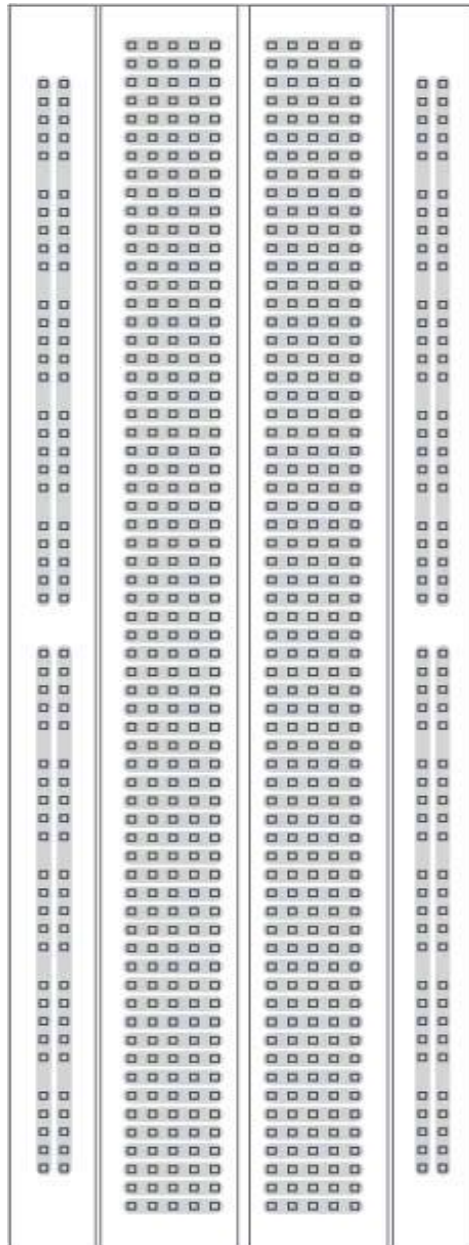


Fig. 4: The metal strips connections of a breadboard

The top and bottom of a breadboard are shown below with the bottom insulation stripped off to clearly show metal strip connections corresponding to the holes. Please note that the orientation of the boards in the diagram below have been rotated by 90° compared to the diagram above.

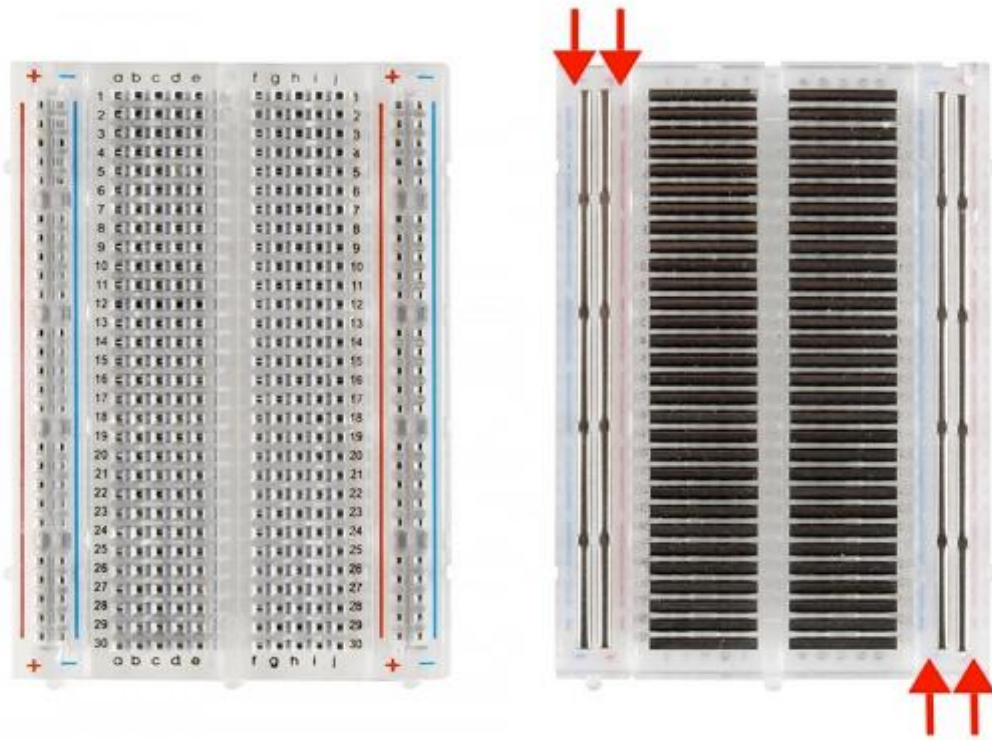


Fig. 5: The top and bottom level of a breadboard

Note how all holes in the selected row are connected together, so the holes in the selected column. The set of connected holes can be called a node:

To interconnect the selected row (node A) and column (node B) a cable going from any hole in the row to any hole in the column is needed:

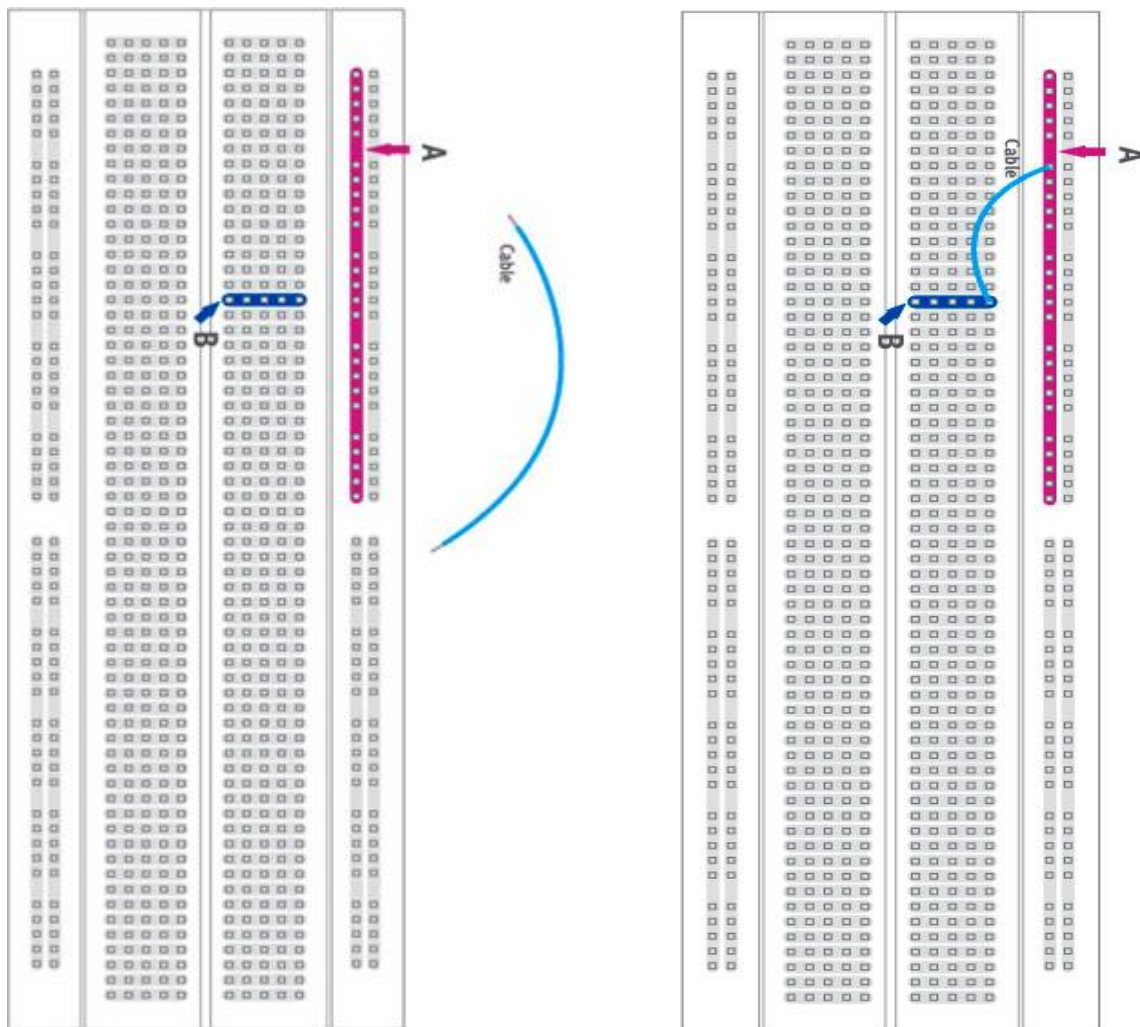


Fig. 7: Different nodes connection on a breadboard

Now the selected column (node B) and row (node A) are interconnected:

1.6 From electronic diagrams to actual circuit connections

(<http://wiring.org.co/learning/tutorials/diagrams/index.html>)

- i. Breadboard
- ii. LED
- iii. 1 k Ω Resistor
- iv. Digital Multimeter (DMM)

A circuit diagram makes use of standardized symbols that represent electrical components or devices. It is easier to draw these symbols than drawing the actual pictures of the components. The actual components might change appearance as the electronics industry revises them or renders them obsolete. The diagrams describe the way in which the components are connected together electrically. There are drawn lines that represent wires or conductors between the appropriate connection points on the symbols; no particular type of wire or physical distance between components is implied; two components might be separated by a few inches or centimeters or a meter or feet.

The following tutorial translates from a circuit diagram to actually connecting components on a breadboard. Note that the circuit diagrams are the universal way of representing circuits; books, on-line resources, and materials use them to communicate the circuit connections. They are very useful compared with pictorial diagrams of the connections.

Let us consider the following circuit diagram:

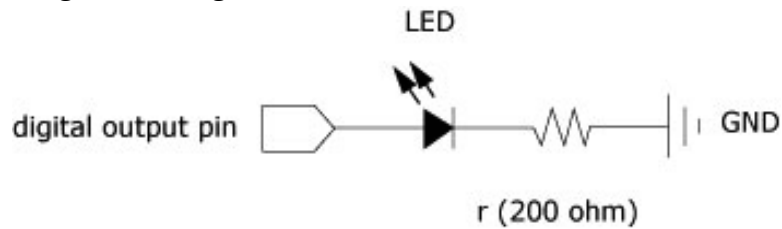


Fig. 8: Simple DC Circuit

The next step would be to identify the components and their terminals:

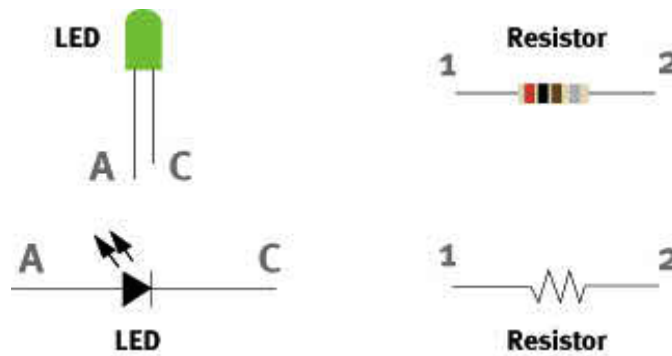


Fig. 9: Circuit Components and their terminals

Next, identify the connection nodes between components, connections between different components are formed by putting their legs (or terminals) in a common node:

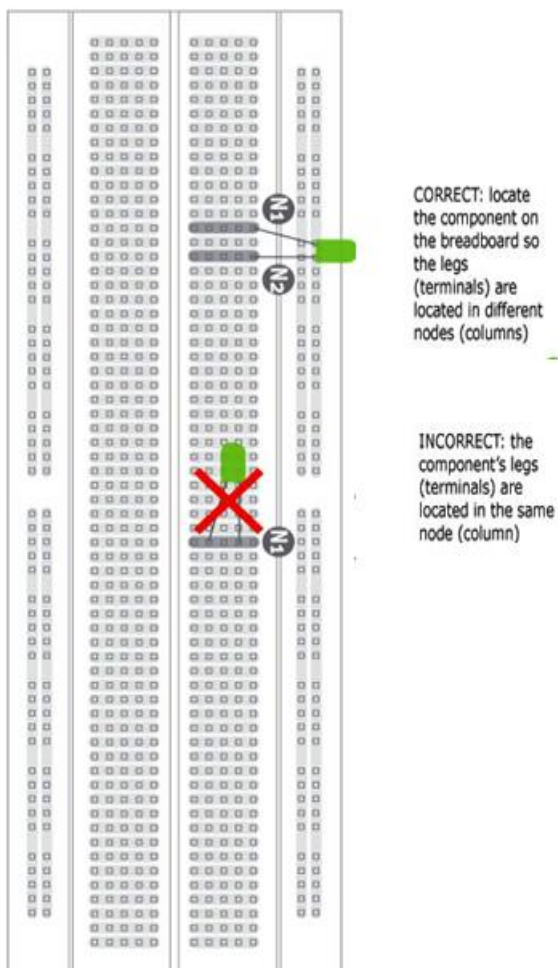


Fig. 10: Component Connections process on breadboard

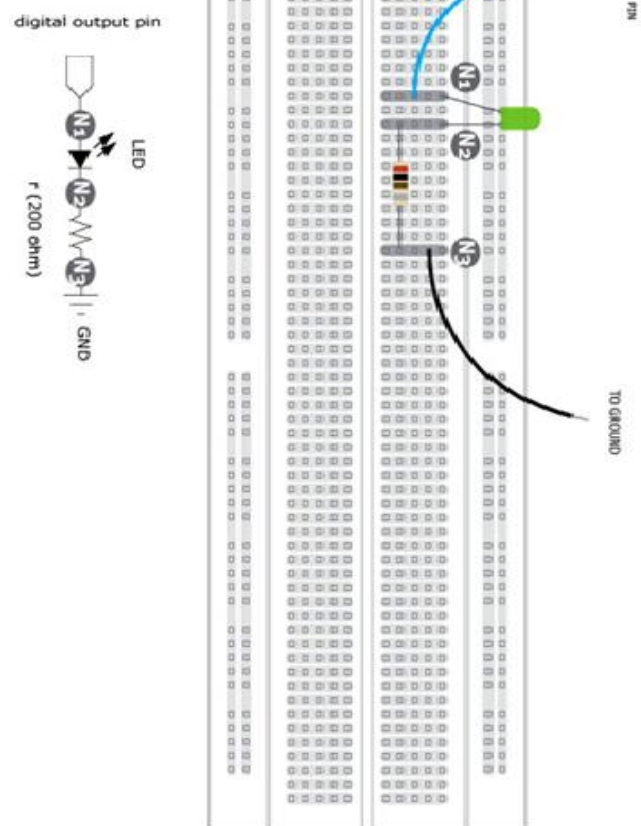


Fig. 11: Full Circuit Connections process on breadboard

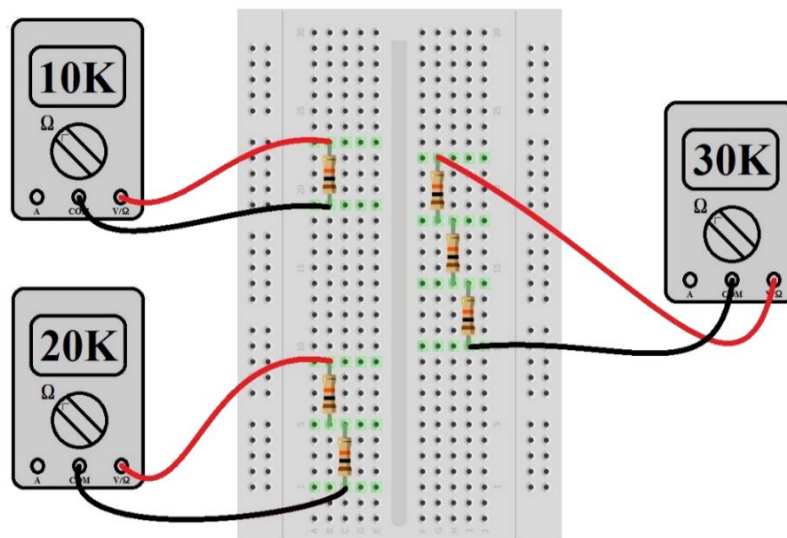
Note the difference between the correct and incorrect connections. In the correct version the two legs are on different columns (nodes), in the incorrect version the two legs are connected to the same column (node) which is equivalent to solder or tie together the two legs of the LED.

The LED has two legs, from Fig 7 the leg marked as A is connected to Node N1, the leg marked C is connected to the leg marked 1 on the resistor (Node N2) and the leg marked 2 on the resistor is connected to GROUND (Node N3). The LED is a polarized device, which means it matters the way it is connected, the resistor is not polarized so pins can be inverted with no effect on the circuit's behavior. To learn more about a specific component try to find its datasheet. Search on the Web using the component's reference number to become familiar with its functions, terminals and specs.

Equivalent resistance:

Source: <https://learn.sparkfun.com/tutorials/series-and-parallel-circuits>

Series:



Parallel:

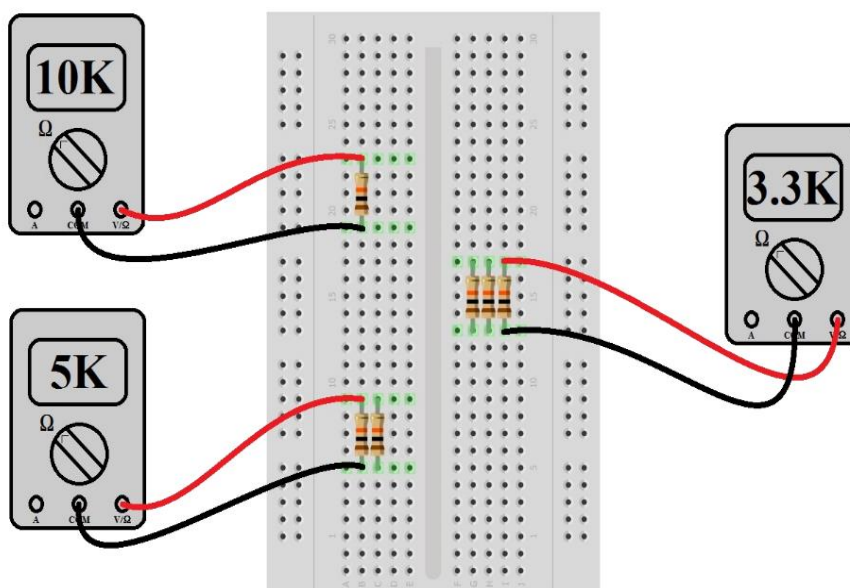


Fig. 12: Resistors Connections process on breadboard

1.7 DC Power Supply:

A DC power supply is a type of power supply that gives direct current (DC) voltage to power a device. Because DC power supply is commonly used on an engineer's or technician's bench for a ton of power tests, they are also often called a "bench power supply."



Fig 13: DC Power Supply

DC Power Supply Front Panel:

- A) Display
- B) Voltage and Current Control
- C) Outputs
- D) Series-Parallel Control
- E) C.C. and C.V. Indicator

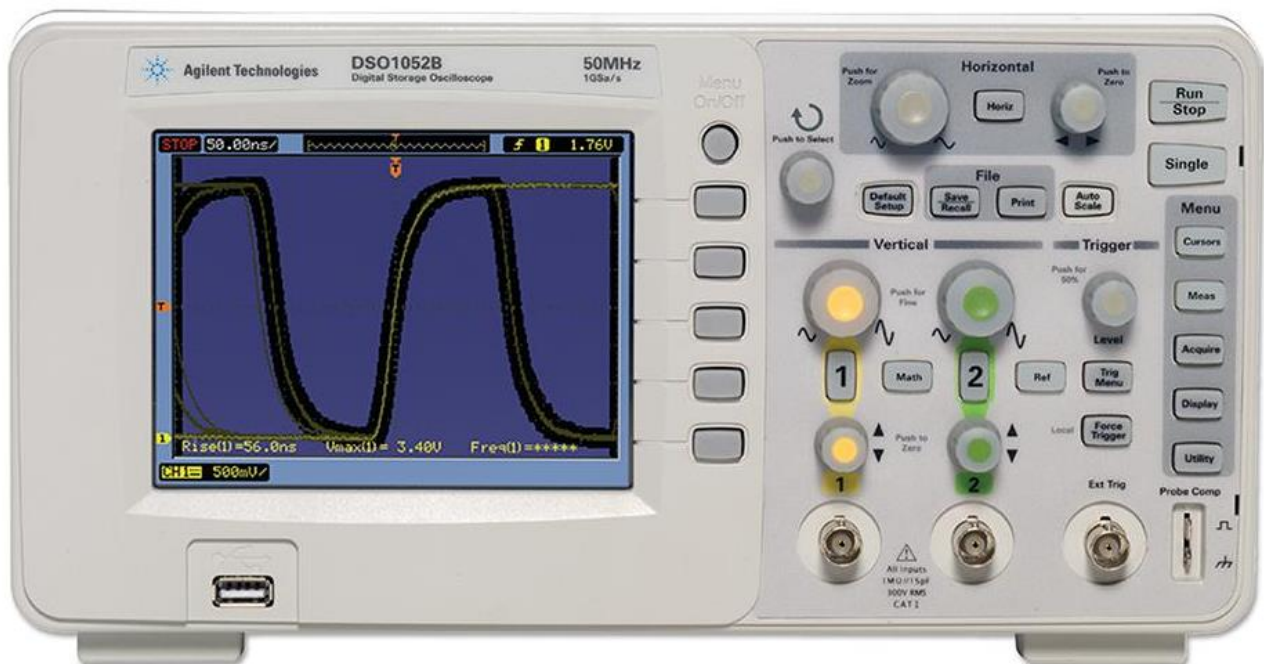
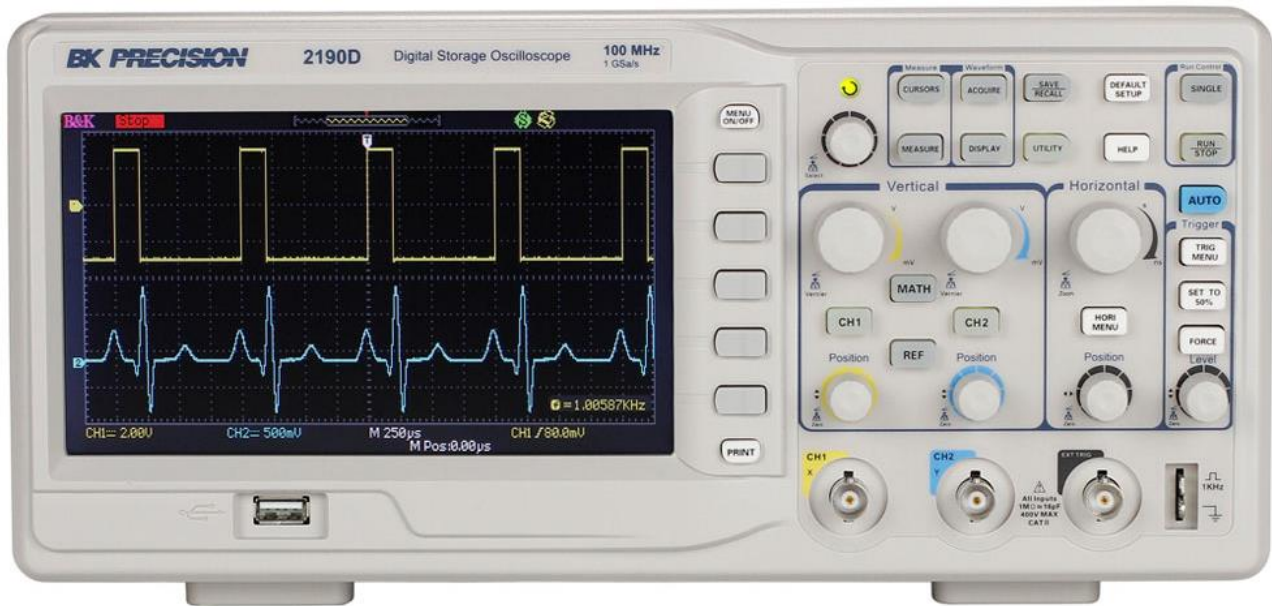
1.8 AC Source / Function Generator / Signal Generator



Function Generator Front Panel:

- A) Frequency Control (Knob and Range)
- B) Amplitude Control
- C) Waveform Control (Sinusoidal, Triangular, Square Waves)
- D) DC Offset Control
- E) Output
- F) Additional Features

1.9 Oscilloscope:

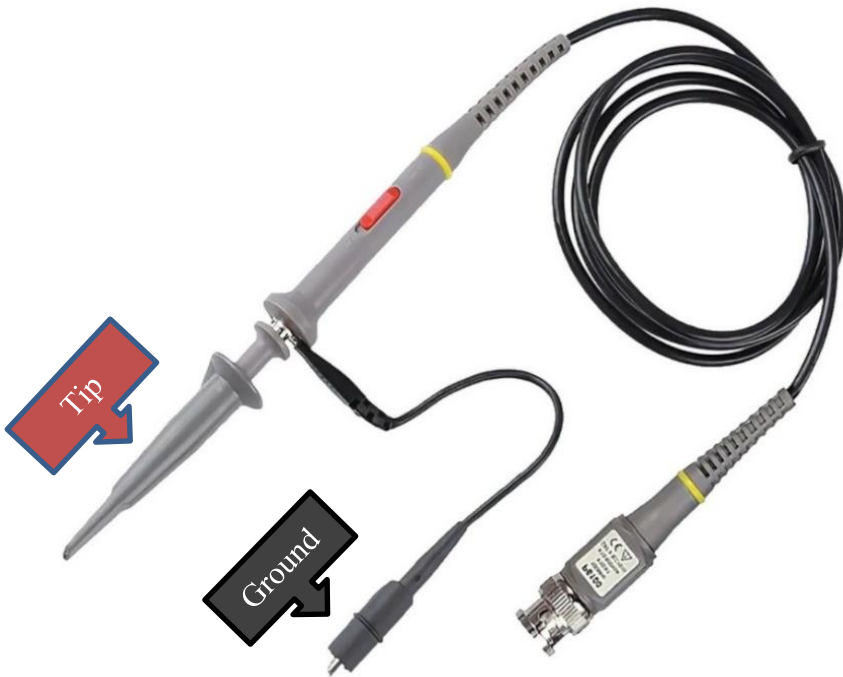


Oscilloscope Front Panel Sections:

- A) Display
- B) Soft buttons and Selection Knob
- C) Vertical Control (Voltage)
- D) Horizontal Control (Time)
- E) Trigger Control
- F) Menu Keys
- G) Training Signal Source
- H) Inputs (Channel 1, Channel 2, External Trigger)

Oscilloscope's Probe:

- Tip
- Ground
- 1X – 10X Slider



Oscilloscope Testing and Calibration:

- Connect Channel 1 and Channel 2 Probes to Training Signal Source
- Press “AUTO” or “AUTO SET”
- Vertical Control (for Voltage)
 - Scale (Volt/Div)
 - Position (Rotate and Press)
- Horizontal Control (for Time)
 - Scale (Second/Div)
 - Position (Rotate and Press)
- Trigger Control
- Function Menu Keys
 - Cursor
 - Measure

Oscilloscope's Display:

